SECTION 1: OBSERVATION FIELD DECLARATION

The Phenomenon: The global response to antimicrobial resistance (AMR), analyzed from the launch of the WHO Global Action Plan (2015) to the present (2026). The phenomenon is not AMR itself as a biological process, but the *human response* to it—the policies, investments, and international declarations that constitute the global attempt to govern this health threat.

Temporal Frame: 2015–2026. This period marks a shift from agenda-setting (the WHO GAP) to a phase of implementation, evaluation, and observed divergence in national and global strategies. It allows for the inclusion of post-pandemic data and an assessment of long-term trends in investment and policy.

Geographic Scope: Global, with necessary focus on key actors: OECD nations, major agricultural producers (e.g., Brazil, China, India, USA), and low- and middle-income countries (LMICs) with high infectious disease burdens and developing health systems.

Why This Phenomenon Produces Interpretive Divergence: AMR is a classic "wicked problem." It is a slow-moving, invisible crisis characterized by deep uncertainties, conflicting interests, and trade-offs that play out across different scales (individual vs. society; present vs. future; human health vs. economic productivity). While the nominal goal of "tackling AMR" is universally agreed upon, the causal attribution of the problem and the locus of responsibility for the solution are radically contested. Is it a market failure? A tragedy of the commons? A symptom of unsustainable development? Each answer leads to a different policy prescription.

Signal Contestation Level: Shared-Signal Domain, but with Contested Ontology. The core signals (death estimates, consumption data, pipeline status) are cited by all major actors. There is little debate over their empirical validity. However, their ontological status—*what they are evidence of*—is deeply contested. For one grammar, a high AMR death toll is evidence of a *market failure* (lack of new drugs). For another, it is evidence of a *health systems failure* (poor stewardship and infection control). This makes the domain a prime candidate for prismatic analysis, as the divergence is not about the facts, but about the interpretive grammar applied to them.

The Major Interpretive Communities:

1. **The Biomedical Innovation Community:** Primarily pharmaceutical industries, venture capitalists, and research-funding bodies. They focus on the technological deficit.
2. **The Global Public Health Community:** WHO, national public health agencies, clinicians, and stewardship advocates. They focus on the appropriate use of existing tools.
3. **The One-Health/Structural Community:** The FAO, UNEP, veterinary associations, and civil society groups focused on social and environmental determinants. They focus on the systemic drivers across human, animal, and environmental sectors.

These three communities form the basis for the grammars to be specified.

SECTION 2: SIGNAL SET

Core Signals (10–15)

Signal Name	Source / Type	Ambiguity / Contestation Note

1. Global AMR Deaths	**Source:** GRAM Project (2019, Lancet) **Type:** Epidemiological estimate	**Ambiguity:** The headline numbers (1.27M direct, 4.95M associated) are widely cited, but the distinction between "attributable" and "associated" leaves room for causal interpretation. Does the death count prove the scale of the problem, or the failure of health systems to manage infection?
2. Human Antibiotic Consumption	**Source:** WHO, IQVIA **Type:** Sales/usage data	**Ambiguity:** Rising consumption can signal *improved access* in LMICs or *overuse/misuse* in high-income countries. The same data point tells two opposing stories depending on context and grammar.
3. Agricultural Antibiotic Use	**Source:** WOAH,各国 government reports **Type:** Sales data, biomass-adjusted usage	**Ambiguity:** Trends show declines in Europe but rises in emerging economies (Brazil, India, China). This creates a tension between celebrating progress in one region and flagging an emerging crisis in another.
4. New Antibiotic Pipeline	**Source:** WHO (Antibacterial Agents in Clinical Pipeline), PEW Charitable Trusts **Type:** R&D data	**Ambiguity:** The pipeline is consistently described as "dry" or "insufficient." However, it contains many chemically modified versions of existing classes (derivatives) and very few novel classes or those targeting critical pathogens. The *quality* of the signal is contested.
5. Market Withdrawals	**Source:** Pharmaceutical industry reports **Type:** Economic/regulatory data	**Ambiguity:** Companies like Novartis, AstraZeneca, and Sanofi have exited or scaled back AMR R&D, citing poor return on investment. This is a clear signal of market failure, but it also shows that the existing pipeline data (Signal 4) is fragile, as late-stage assets lack a commercial home.
6. Stewardship Programme Implementation	**Source:** WHO (AMR country self-assessment survey - TrACSS) **Type:** Policy self-reporting	**Ambiguity:** High-income countries report high implementation rates, but this measures policy adoption, not clinical behaviour change. The data is self-reported and prone to optimistic bias, masking variability in on-the-ground enforcement.
7. WHO/OECD Economic Modelling	**Source:** OECD (Stemming the Superbug Tide) **Type:** Cost-benefit projections	**Ambiguity:** Models project massive GDP losses (e.g., 3.8% annual loss by 2050) and healthcare costs. The models are powerful advocacy tools, but their long-term projections are sensitive to assumptions about discount rates and the pace of technological change.

8. One-Health Framework Adoption	**Source:** National Action Plans (NAPs), Tripartite (WHO/FAO/WOAH) assessments **Type:** Policy analysis	**Ambiguity:** Most NAPs claim to be "One-Health," but the reality is often a siloed approach. The signal is performative—declared intent versus integrated action—making it difficult to quantify genuine adoption.
9. Surveillance Data Quality	**Source:** GLASS (Global Antimicrobial Resistance and Use Surveillance System) **Type:** Public health data	**Ambiguity:** A deep divide exists. High-income countries provide high-quality, granular data (e.g., on specific pathogen-drug combinations). LMICs often provide no data or only aggregate reports. The map of AMR is, therefore, a map of surveillance capacity, creating a blind spot where the burden may be highest.
10. Pharmaceutical Investment Patterns	**Source:** Venture capital reports, public funding announcements (e.g., CARB-X, GARDP) **Type:** Economic/funding data	**Ambiguity:** While big pharma has largely retreated, public-private partnerships and biotechs have emerged. This signals a shift in the *model* of innovation, from blockbuster drugs to a more socially-driven, de-risked model. The grammar determines whether this is seen as a lifeline or a sticking plaster.
11. Diagnostic Test Uptake	**Source:** Clinical studies, market research **Type:** Usage/implementation data	**Ambiguity:** Rapid diagnostics are hailed as a key stewardship tool, but their uptake is low globally. This can be framed as a *cost* barrier (an economic problem) or a *clinical workflow* barrier (a behavioural/systems problem).
12. National Action Plan (NAP) Funding	**Source:** WHO, World Bank **Type:** Budget analysis	**Ambiguity:** Most LMICs have NAPs, but they are almost universally underfunded and reliant on external donors. The existence of a plan (a policy success) contrasts with the lack of implementation capacity (an operational failure).

Counter-Signals (3–5)

Counter-Signal Name	Source / Type	What it challenges
CS1: Declining Agricultural Use (EU)	**Source:** European Medicines Agency **Type:** Sales data	Challenges the narrative of *universally* rising or unmanaged agricultural use. It shows that policy interventions (e.g., the EU ban on metaphylactic use) can work in specific contexts.
CS2: Novel Antibiotic Approvals (2020-2025)	**Source:** FDA, EMA **Type:** Regulatory data	Challenges the "dry pipeline" narrative. A handful of new agents (e.g., cefiderocol, some novel beta-lactamase inhibitor combinations) have been approved, suggesting the pipeline is not completely dry, even if imperfect.

CS3: Post-Pandemic Consumption Dip	**Source:** IQVIA **Type:** Sales/usage data	Challenges the narrative of inexorable consumption growth. Reduced respiratory infections during COVID-19 lockdowns (2020-2021) led to a temporary but significant global drop in antibiotic use, showing that demand is elastic and linked to other factors.
CS4: AMR Awareness in LMIC Communities	**Source:** Qualitative studies (e.g., ReAct, IDRC) **Type:** Ethnographic research	Challenges the top-down narrative that lack of awareness is the primary driver of misuse. Studies show communities often understand antibiotic ineffectiveness intuitively but are trapped by poor access to quality healthcare and diagnostics.

Contested Signals (0–5)

Contested Signal	Description of the Signal Itself	Interpretation in Grammar A	Interpretation in Grammar B	Interpretation in Grammar C
CON1: "Access" vs. "Excess" Consumption	The signal is the *meaning* of a given increase in per-capita antibiotic consumption.	**Grammar A:** Increased consumption, especially in growing economies, signals a *market opportunity* and a failure of *access* to newer, more effective (and often patented) drugs.	**Grammar B:** Increased consumption signals a stewardship failure—an *excess* of inappropriate use that drives resistance, regardless of the drug's age.	**Grammar C:** The signal is an index of *structural inequality*. It is "access" in places without clean water and functioning clinics, and "excess" in industrial farming and wealthy healthcare systems. The signal itself is not one thing; it's two different phenomena happening simultaneously.

SECTION 3: GRAMMAR SPECIFICATIONS

Grammar A: The Market-Failure/Innovation Grammar

- **Source Tradition:** Neoclassical economics, pharmaceutical industry strategy, venture capital.
- **Primary Analytical Variable: Return on Investment (ROI).** The fundamental problem is the broken market for antibiotics. Low prices (due to stewardship/preservation) and short treatment durations make them commercially unattractive compared to chronic disease drugs.
- **Signal Selection Rules:** Prioritize signals that indicate economic viability: pipeline data, market withdrawals, investment patterns, and the economic costs of AMR (as potential market size). Signals of stewardship are less relevant unless they impact future market attractiveness.
- **Weighting Priority:**
    1. **Highest:** Pipeline data, market withdrawals, investment patterns.
    2. **Medium:** AMR death tolls (as unmet medical need), economic modelling of GDP loss.

    3. **Lowest:** Stewardship implementation rates, agricultural use patterns, One-Health framework adoption (seen as peripheral to the core problem of innovation).
- **Causal Chain Type: Linear-Economic.** Broken market incentives → Disinvestment by pharma → Dry pipeline → Lack of new drugs → Increased mortality and economic loss. The solution is to fix the incentive (the cause).
- **Admissible Closure:** The solution must involve "pull" incentives (e.g., market entry rewards, transferable exclusivity vouchers) to delink profitability from volume of sales. Push funding (e.g., CARB-X) is a necessary but insufficient first step.
- **Excluded Signals:** Qualitative community studies (CS4) are excluded as anecdotal. The "Access vs. Excess" ambiguity (CON1) is resolved by defaulting to "access" as the primary frame for consumption growth.
- **Characteristic Vocabulary:** "Market failure," "broken pipeline," "pull incentives," "de-linking," "innovation ecosystem," "unmet medical need," "economic burden."
- **Temporal Orientation: Forward-looking/Prescriptive.** The grammar is focused on building a future state where the market for antibiotics functions like markets for other drugs.

Grammar B: The Public Health/Stewardship Grammar

- **Source Tradition:** Clinical medicine, epidemiology, WHO, public health agencies.
- **Primary Analytical Variable: Selective Pressure.** The fundamental problem is the overuse and misuse of antibiotics in human healthcare, which drives the evolution and spread of resistance. The existing arsenal is a precious, finite resource to be conserved.
- **Signal Selection Rules:** Prioritize signals that indicate usage patterns and their consequences: human consumption data, stewardship implementation, surveillance data (to track resistance resulting from use), and the direct health burden of AMR (deaths, infections).
- **Weighting Priority:**
    1. **Highest:** Global AMR deaths, stewardship implementation rates, human consumption trends, surveillance data quality.
    2. **Medium:** Agricultural use (as a source of resistant organisms affecting humans), diagnostic uptake.
    3. **Lowest:** Pipeline data and investment patterns (seen as a distant, secondary concern; new drugs will just be misused if stewardship isn't fixed first).
- **Causal Chain Type: Cyclical-Biological.** Overuse (in humans) → Selection pressure → Resistance → Treatment failure → Increased mortality → (attempt to) Use more or newer drugs → (repeat cycle). The solution is to break the cycle at the point of use.
- **Admissible Closure:** The solution must focus on optimizing the use of existing antibiotics through stringent stewardship programs, infection prevention and control (IPC), and better diagnostics. New drugs are welcome but must be protected by strict stewardship from day one.
- **Excluded Signals:** Pharmaceutical investment patterns are noted but treated as exogenous—a problem for another department. The economic logic of ROI is acknowledged but subordinated to the clinical logic of conservation.
- **Characteristic Vocabulary:** "Antimicrobial stewardship," "preserve the miracle," "appropriate use," "infection prevention and control," "treatment guidelines," "resistance patterns," "conservation."
- **Temporal Orientation: Present-focused/Prescriptive.** The grammar is focused on managing the here-and-now crisis in hospitals and clinics, preserving the utility of today's drugs.

Grammar C: The One-Health/Structural Grammar

- **Source Tradition:** Ecological economics, veterinary medicine, environmental science, political ecology, FAO, UNEP.
- **Primary Analytical Variable: Systemic Metabolic Flow.** The fundamental problem is the scale of antibiotic throughput across the entire human-animal-environmental system. It's an issue of

industrial overproduction and waste, not just misuse. Resistance is a pollutant, and antibiotics are an industrial input.

- **Signal Selection Rules:** Prioritize signals that trace the flow of antibiotics through systems: agricultural use (total volume), environmental contamination (largely absent from the provided set, but a key gap), One-Health framework adoption, and the structural drivers of consumption (e.g., intensive farming practices).
- **Weighting Priority:**
    1. **Highest:** Agricultural antibiotic use (total volume), One-Health adoption (as a marker of systemic thinking), the "Access vs. Excess" signal (CON1) as the central interpretive problem.
    2. **Medium:** Human consumption trends, surveillance data quality (especially its gaps in LMICs and environmental sectors), economic modelling (as a tool to show systemic costs).
    3. **Lowest:** Pipeline data, market withdrawals, stewardship in high-income hospitals (seen as optimizing a fundamentally broken, high-throughput system).
- **Causal Chain Type: Complex Systems.** Industrial agriculture's reliance on antibiotics for growth promotion/metaphylaxis → Environmental contamination with antibiotics and resistant genes → Dissemination of resistance through water, soil, and food chains → Human exposure (community, not just hospital) → Health burden. This is a non-linear, multi-causal web.
- **Admissible Closure:** The solution must be systemic and regulatory: drastically reducing agricultural use through bans, investing in water and sanitation (WASH) to break environmental transmission, and rethinking industrial food production models. Innovation is relevant only if it creates alternatives to antibiotics (e.g., vaccines, probiotics, phage therapy).
- **Excluded Signals:** The specific ROI calculations for new drugs (Grammar A) are seen as a detail within a much larger and more destructive economic system. Stewardship in high-income hospitals is "end-of-pipe" solutions.
- **Characteristic Vocabulary:** "One Health," "environmental reservoir," "antimicrobial pollution," "food systems," "intensive agriculture," "metaphylaxis," "WASH," "systemic driver," "ecological perspective."
- **Temporal Orientation: Lateral/Hermeneutic.** The grammar looks across sectors to map the complex interconnections that other grammars miss, revealing the hidden flows that constitute the problem.

Grammar Boundaries:

- **A → B:** Grammar A (Innovation) transforms into Grammar B (Stewardship) when the focus shifts from *creating new tools* to *managing the use of existing ones*. This happens when a new drug is approved; it immediately passes from the jurisdiction of the innovator (A) to the regulator and clinician (B). The boundary is the clinic door.
- **B → C:** Grammar B (Stewardship) transforms into Grammar C (Structural) when the unit of analysis expands from the *individual patient/clinic* to the *population/food system*. This happens when an outbreak of a resistant infection is traced not to a hospital but to a factory farm or a contaminated river downstream from a pharmaceutical plant. The boundary is the hospital walls.
- **C → A:** Grammar C (Structural) could transform into Grammar A if the solution to the systemic flow of pollution is framed as a technological fix (e.g., developing a probiotic that replaces antibiotics in feed, creating a new market). The boundary is the point where a systemic problem is redefined as a technological opportunity.

SECTION 4: SIGNAL PROCESSING

Grammar A Processing: The Market-Failure/Innovation Grammar

We begin with the primary analytical variable: **Return on Investment (ROI)** . This is the lens through which all signals are refracted. The story of AMR, from this vantage, is a story of economic misalignment.

The **Global AMR Deaths (Signal 1)** —1.27 million directly attributable—is processed first. This is not primarily a tragedy of suffering, but an index of an enormous, unmet **market opportunity**. These are potential customers for whom the current standard of care (older, generic antibiotics) is failing. The **WHO/OECD Economic Modelling (Signal 7)** , projecting trillions in GDP loss, reinforces this by quantifying the macroeconomic *cost of inaction*, which is the mirror image of the potential *value of action* (i.e., a new, effective drug).

This lens immediately illuminates the **New Antibiotic Pipeline (Signal 4)** and **Market Withdrawals (Signal 5)** . The pipeline isn't just "dry"; it's a landscape of failed business models. The presence of mostly derivatives is evidence of low-risk, low-reward R&D, a symptom of an incentive system that doesn't reward true innovation. The **Market Withdrawals** by major pharma companies are the most damning signals. They are not acts of corporate irresponsibility but rational responses to a broken market. When a CEO sees a **Counter-Signal like CS2 (Novel Antibiotic Approvals)** , the grammar does not celebrate. It asks: "Who developed these, and are they commercially sustainable?" Most novel approvals come from biotechs or with heavy public subsidy—evidence that the *model* is still broken. The approvals are exceptions that prove the rule.

Human Antibiotic Consumption (Signal 2) is viewed with ambivalence. In wealthy nations with high stewardship, flat or declining consumption is a worry—it shrinks the potential market for new, premium-priced antibiotics that need to be used *less* to preserve them. This is the core paradox the grammar grapples with. The ideal new antibiotic, from a public health and commercial standpoint, is one that is highly effective, prescribed sparingly, and commands a very high price per course. This is a difficult but not impossible business model. However, rising consumption in LMICs (Signal 2) is seen more positively. It signals **access** (CON1) to healthcare and growing economies where newer, patented drugs could potentially find a market, provided intellectual property is respected.

Counter-Signal CS1 (Declining Agricultural Use in the EU) is noted but deemed irrelevant. Agriculture is a massive, low-price, high-volume market for generics, not the premium market Grammar A cares about. It has little bearing on the ROI for a novel, hospital-only Gram-negative antibiotic. Similarly, **Agricultural Antibiotic Use (Signal 3)** in Brazil and China is a non-signal for innovation, though it may be a signal for future resistance that will create *future* unmet need.

Pharmaceutical Investment Patterns (Signal 10) are the grammar's lifeblood. The retreat of Big Pharma is the central problem. The emergence of public-private partnerships like CARB-X and the GAIN Act in the US are processed as partial, "push" solutions—they fund the risky early science, but they don't solve the "pull" problem. The grammar's weight is placed on the lack of viable exit strategies for these biotechs. Without a clear path to profitability (through acquisition or sales), the entire innovation ecosystem stalls. This is where the causal chain is anchored: broken incentives (no pull) → no ROI → no investment → no new drugs.

The grammar must confront a powerful **counter-signal tension** here. The very logic of stewardship (Signal 6) is at odds with its own. **Stewardship Programme Implementation (Signal 6)** , from a clinical perspective, aims to *limit* the use of new, powerful antibiotics. For Grammar A, this is a direct threat to ROI. The admission of this tension is processed by arguing that new economic models, like the UK's Netflix-style subscription, can "de-link" profit from volume. The grammar's solution, therefore, is to make stewardship *economically neutral* for the innovator. The state pays a fixed fee for access to the drug, regardless of how much it's used. This transforms stewardship from a threat into a feature.

The **Admissible Closure** is now inescapable. The primary intervention must be large-scale, sustainable "pull" incentives. Push funding is a down payment. Without a market entry reward of $1-3 billion per successful drug, or a transferable exclusivity voucher that can be sold, the pipeline will remain dry. The

Surveillance Data Quality (Signal 9) divide matters only insofar as it helps define where a future drug's premium market might be—in countries with the health systems to pay for it.

(This closure is a product of processing the signals through a grammar that treats economic incentives as the fundamental causal variable. It is not a statement of fact, but an architectural output.)

Grammar B Processing: The Public Health/Stewardship Grammar

We begin with the primary analytical variable: **Selective Pressure.** The fundamental driver of AMR is the volume and appropriateness of antibiotic use. The crisis is one of *consumption*, not production.

The **Global AMR Deaths (Signal 1)** are processed immediately and directly. They are not a market signal but a **clinical audit**. Each death represents a failure of the current system to either prevent the infection (IPC) or treat it effectively because the drugs no longer work. This is the tragic outcome of decades of applying relentless selective pressure. The deaths validate the urgency of the stewardship mission.

The **Human Antibiotic Consumption (Signal 2)** is the primary vital sign. Rising global consumption is a direct measure of increasing threat. The grammar's first instinct is to disaggregate this signal. It zeroes in on **Contested Signal CON1 (Access vs. Excess)** . For Grammar B, while acknowledging the access problem, the dominant framing must be "excess." The sheer volume of inappropriate prescribing for viral infections (colds, flu) in both high- and low-income settings is the key driver. This is where stewardship programmes have the most leverage. **Stewardship Programme Implementation Rates (Signal 6)** are therefore the primary metric of *response*. The gap between policy (reported in TrACSS) and practice is the central frustration. The grammar processes the high self-reported rates from OECD nations with skepticism, looking instead for evidence of behavioural change, such as **Diagnostic Test Uptake (Signal 11)** . Low uptake of rapid diagnostics, even in wealthy countries, is a critical counter-signal to claims of stewardship success. It shows that the clinical workflow and habits remain unchanged.

Agricultural Antibiotic Use (Signal 3) is a major concern. While human medicine is the direct cause of resistance in human pathogens, agriculture acts as a massive, parallel reservoir of selective pressure. The grammarians here would process the data on rising use in Brazil and India as an alarming source of resistance genes that can flow back into human populations via food and the environment. They would note **Counter-Signal CS1 (Declining EU Use)** as proof that policy intervention works and should be replicated globally.

The **New Antibiotic Pipeline (Signal 4)** and **Market Withdrawals (Signal 5)** are viewed through a very specific lens: anxiety. New drugs are not a solution; they are a temporary, fragile resource. The pipeline data is processed not with excitement but with concern. "Will we have enough new drugs to replace the ones we're losing?" is the wrong question. The right question is, "How can we ensure any new drug is not rapidly rendered useless by the same patterns of misuse that destroyed the utility of its predecessors?" **Market withdrawals** by pharma are met with a shrug. The grammar's tradition has a long memory; it recalls that nearly all important antibiotic classes were discovered in a "golden age" that ended decades ago. It does not expect or rely on industry to save the day. The **Counter-Signal of Novel Approvals (CS2)** is met with a demand: impose strict stewardship on these new agents from day one.

This creates an internal **counterpoint**. The grammar must reconcile its own logic: it needs new drugs to treat the hardest cases (e.g., CRE infections), but it fears that their very existence will lead to overuse and the rapid erosion of their effectiveness. The solution is to place new drugs in a protected category, with restricted use overseen by infectious disease specialists and prior authorization. This is the grammar's core closure: control.

The **WHO/OECD Economic Modelling (Signal 7)** is a useful advocacy tool. The projected GDP losses are powerful for convincing finance ministries to fund stewardship programmes and IPC. But the grammar is

ultimately driven by health outcomes, not economic ones. The high cost of a single resistant infection (prolonged ICU stay, isolation, etc.) is a more immediate and compelling signal.

The **One-Health Framework Adoption (Signal 8)** is promising but largely performative. Grammar B sees the need for coordination between human and vet medicine but often defaults to a human-centric view. The environmental component is acknowledged but seen as a downstream consequence, not an upstream driver. The grammar's **Excluded Signal** is the deep, qualitative understanding of why people misuse antibiotics (CS4). It tends to assume that misuse stems from a lack of knowledge (an educational deficit), rather than from a lack of access to functioning health systems (a structural deficit). The solution, therefore, is more education and more guidelines for clinicians and patients.

The **Admissible Closure** is a suite of interventions focused on the point of care: universal implementation of hospital stewardship programmes with dedicated pharmacists and ID physicians, public campaigns to reduce demand for antibiotics for self-limiting illnesses, massive investment in IPC (hand hygiene, sanitation), and rapid diagnostic tests at the point of care. The goal is to preserve the utility of the existing arsenal for as long as possible. New drugs are a last resort, to be kept in a locked cabinet for emergencies.

(This closure is a product of processing the signals through a grammar that treats biological selection pressure and clinical practice as the fundamental causal architecture.)

Grammar C Processing: The One-Health/Structural Grammar

We begin with the primary analytical variable: **Systemic Metabolic Flow.** The problem is not a broken market nor a clinical bad habit; it is the industrial-scale throughput of antibiotics through the global economy. They are an invisible pollutant, and resistance is the resulting ecological disruption.

The most important signal is **Agricultural Antibiotic Use (Signal 3)** , specifically the total tonnage used globally, which dwarfs human consumption. This is not just a "contributing factor"; it is the engine room of the problem. The grammar processes the rise in use in Brazil, China, and India not as a national policy failure but as a direct consequence of the global industrial food system's demand for cheap meat. Intensive animal agriculture, with its reliance on metaphylaxis (treating entire flocks/herds) and historical use of growth promoters, is the primary site of antibiotic flow. **Counter-Signal CS1 (Declining EU Use)** is processed as a success, but one that highlights the failure to export those regulatory standards globally. It shows that the flow *can* be staunched, but only with political will.

Human Antibiotic Consumption (Signal 2) is immediately refracted through **Contested Signal CON1 (Access vs. Excess)** . This grammar treats the ambiguity not as a problem to be resolved, but as the core diagnostic. It sees both simultaneously: the "excess" of high-income, high-consumption healthcare and agriculture, and the tragic "access" deficit in LMICs where people die from common bacterial infections because they cannot get any antibiotic at all. The true crisis is this structural inequality embedded within a high-throughput system.

The **Global AMR Deaths (Signal 1)** are read as a map of these inequalities. The 4.95 million associated deaths are likely a vast underestimate because **Surveillance Data Quality (Signal 9)** is poorest in precisely the places where the burden is highest—LMICs with poor WASH (Water, Sanitation, Hygiene). The lack of data from these regions is itself the most damning signal. It reveals a global blind spot created by a surveillance system designed for wealthy hospitals, not for communities where transmission occurs through contaminated water and food.

This brings the environmental dimension into focus. The signal set provided lacks direct data on antibiotic residues in the environment, but Grammar C would infer it. From the high agricultural use, it deduces massive excretion of active compounds and resistant genes into soil and water systems. These become environmental reservoirs, circulating back into human populations. **One-Health Framework Adoption (Signal 8)** is the only signal that gestures towards this complexity. However, the grammar processes its

adoption with deep skepticism. Most NAPs claim "One Health," but Grammar C sees this as a "siloed" approach where human, animal, and environment ministries still operate independently. True One-Health would mean the environment ministry having veto power over agricultural antibiotic policy.

The **New Antibiotic Pipeline (Signal 4)** and **Pharmaceutical Investment Patterns (Signal 10)** are processed as a dangerous distraction. Focusing on new drugs is like trying to solve a river pollution crisis by distributing more bottled water. It ignores the source. Grammar C would point out that the **Market Withdrawals (Signal 5)** of big pharma are a symptom of the same industrial logic that created factory farming—a logic of maximizing throughput and profit, which is fundamentally incompatible with sustainability. The grammar is more interested in alternatives to antibiotics: vaccines for livestock, improved animal husbandry that reduces stress and infection, phage therapy, and probiotics. These are signals not present in the mainstream set, representing a different vision of innovation.

The grammar's internal **counterpoint** is the tension between its radical systemic critique and the need for pragmatic action. The **WHO/OECD Economic Modelling (Signal 7)** is a tool that can be repurposed. Instead of just showing GDP loss, it can be used to model the full social cost of antibiotic use, including environmental damage, which is currently treated as an externality. This could justify strong regulations.

The **Admissible Closure** is not just about better stewardship or new pull incentives. It demands fundamental changes to the global food system: ending the use of medically important antibiotics for growth promotion and routine disease prevention in healthy animals; investing in WASH infrastructure as a core AMR intervention; and regulating pharmaceutical waste from manufacturing. The focus shifts from the clinic to the farm and the factory. The solution is to reduce the total flow of antibiotics through the system, thereby lowering the background selective pressure on a global scale.

(This closure is a product of processing the signals through a grammar that treats the global industrial system and its material flows as the fundamental causal architecture.)

SECTION 5: CLOSURE COMPARISON

Dimension	Grammar A: Market-Failure	Grammar B: Public Health/Stewardship	Grammar C: One-Health/Structural
Closure Content	Implement large-scale "pull" incentives (market entry rewards) to fix the broken innovation ecosystem. Push funding is necessary but insufficient.	Implement stringent stewardship, IPC, and diagnostic programmes to optimize use of existing drugs and preserve their efficacy. New drugs are a last resort.	Implement systemic regulations to reduce the total volume of antibiotic use, especially in agriculture, and invest in WASH to break environmental transmission cycles.
Signal Divergence	Pipeline data, market withdrawals, and investment patterns are the most salient. AMR deaths signal market size.	Human consumption, stewardship rates, and surveillance data are most salient. AMR deaths signal clinical failure.	Agricultural use, One-Health adoption, and data gaps are most salient. AMR deaths signal systemic inequality.

Exclusion Patterns	Excludes qualitative data on user behaviour (CS4). Treats the "access vs. excess" signal as primarily "access." Ignores environmental flows.	Excludes deep structural drivers (e.g., industrial farming economics). Treats "access vs. excess" as primarily "excess" to be managed. Sees environment as a secondary concern.	Excludes the specific ROI logic of pharma. Views stewardship as an "end-of-pipe" solution that doesn't address the source.
Causal Architecture	Linear-Economic: Broken incentives → No R&D → No drugs → Deaths.	Cyclical-Biological: Overuse → Resistance → Treatment Failure → (Attempt to) Use more.	Complex Systems: Industrial throughput → Environmental pollution → Widespread resistance → Health burden.
Closure Range	Narrow: Focused on the R&D ecosystem and high-income country markets.	Intermediate: Focused on healthcare settings, primarily in countries with functioning health systems.	Broad: Encompasses agriculture, environment, food systems, and global inequality.
Closure Timing & Confidence	Fast/High Confidence: The economic logic is clear and self-contained. The solution (pull incentives) directly follows from the problem (broken market).	Medium/Moderate Confidence: The problem is urgent and visible, but the solution is difficult and behavioural. Success is hard to measure and sustain.	Slow/Low Confidence: The problem is vast and complex, the proposed solutions are politically and economically challenging, and the causal chains are long and diffuse.

Ontological Check: Do the grammars constitute the same object?

No, they do not.

- For Grammar A, the object of concern is the **Innovation Ecosystem**. AMR is a problem of technological deficit.
- For Grammar B, the object of concern is the **Clinical Arsenal**. AMR is a problem of resource depletion and misuse.
- For Grammar C, the object of concern is the **Industrial Metabolic System**. AMR is a problem of ecological pollution and structural inequality.

They are analysing three different, overlapping objects that share the name "the global response to AMR." Grammar A is trying to build a machine (the innovation pipeline) that Grammar B is trying to protect patients from (by limiting use of its output), while Grammar C argues that the fuel for the machine (the antibiotics) is poisoning the entire ecosystem. The profound divergence is not about which policy is best, but about what the fundamental problem *is*.

SECTION 6: COUNTER-SIGNALS, LEAKAGE, AND STRESS TEST

Consolidated Counter-Signal Analysis:

- **CS1 (Declining EU Ag Use):** Successfully absorbed by all grammars. Grammar A ignores it; B uses it as evidence for policy; C uses it as evidence for the possibility of systemic change, while lamenting its limited scope.
- **CS2 (Novel Antibiotic Approvals):** Creates significant tension for Grammar A (are things really that bad?) and Grammar B (are new drugs a threat or a blessing?). It is absorbed by A by arguing the model is still broken, and by B by demanding strict stewardship for these new agents. It is dismissed by C as a distraction.
- **CS3 (Post-Pandemic Consumption Dip):** Absorbed by all. Grammar A sees it as a market contraction; B sees it as proof of the link between viral illness and antibiotic misuse; C sees it as a glimpse of a low-throughput world.
- **CS4 (AMR Awareness in LMICs):** This is the most challenging counter-signal for Grammar B. It directly challenges its core assumption that misuse stems from ignorance, pointing instead to structural access problems. Grammar B struggles to absorb this without borrowing concepts from Grammar C.

Leakage Instances:

- **Grammar A leaks into B:** When a proponent of market solutions argues that new drugs will be useless without stewardship, they are momentarily adopting Grammar B's logic. The concept of "de-linking" profit from volume is an attempt to build a bridge between A and B.
- **Grammar B leaks into C:** When a hospital stewardship officer realizes that a patient's resistant infection came from a nursing home or a factory-farmed chicken, they are forced to look beyond the clinic walls. The logic of their work "leaks" into the systemic concerns of Grammar C.
- **Grammar C leaks into A:** When an environmentalist argues for the development of a vaccine for livestock to replace antibiotics, they are, for a moment, operating in the innovation-focused space of Grammar A, albeit with a different goal.

Grammar Stress Test:

- **Grammar A (Market-Failure):**
    - **Most Threatening Counter-Signal: CON1 (Access vs. Excess)** framed as overwhelming "Excess." If the primary consumption dynamic is not growing access in LMICs (a potential market) but rampant, low-price overuse of generics everywhere (which destroys the utility of all drugs), then the entire premise of a premium-priced, conserved antibiotic becomes economically unviable at scale. The market for a drug you can't use is not a market.
    - **Can it absorb?** Only with difficulty. It would have to argue that its proposed pull incentives are so large that they override this concern, or that future diagnostic breakthroughs will enable precision use, thus "creating" a market even within a stewardship context. It requires a leap of faith.
- **Grammar B (Public Health/Stewardship):**
    - **Most Threatening Counter-Signal: CS4 (AMR Awareness in LMICs)** combined with **Signal 9 (Surveillance Data Quality)** . If the problem in high-burden settings is not ignorance but a lack of access to clean water, functioning clinics, and reliable diagnostics, then its core toolkit of guidelines and education is almost useless. Stewardship becomes a luxury of the well-resourced health system.
    - **Can it absorb?** Partially, but it requires significant expansion. It must broaden its closure to include massive investments in WASH and primary healthcare infrastructure, moving it closer to Grammar C's territory. If it refuses, its closure is conditional on a context of functional health systems, which is not the global reality.
- **Grammar C (One-Health/Structural):**
    - **Most Threatening Counter-Signal: CS2 (Novel Antibiotic Approvals)** combined with the sheer political inertia of the global food system. If new, powerful drugs continue to be approved, they provide a temporary reprieve that undermines the urgency of its systemic

argument. Politicians and industry can point to these new tools and say, "See? Technology will save us, we don't need to reform industrial agriculture."
    - **Can it absorb?** It would argue that these are temporary palliatives that will be quickly absorbed into the same metabolic flow, creating even more potent forms of pollution (resistance to last-resort drugs). Its closure is conditional on the failure of technological silver bullets. If technology perpetually stays one step ahead, its systemic critique is politically neutered.

SECTION 7: POST-ANALYSIS REFLEXIVE NOTE

- **Was your home grammar prediction correct?** Yes. My initial self-identification was with Grammar B (Public Health/Stewardship). As a product of a public health and social science background, my instinct is to focus on behavioural interventions, health systems, and clinical practice. This analysis confirmed that bias; processing signals through Grammar B felt intuitive and familiar.
- **Which grammar was actually hardest?** Grammar A (Market-Failure) was the hardest to operate faithfully. My personal skepticism towards pharmaceutical industry motives and market-based solutions to health crises created a constant internal resistance. I had to consciously suppress the urge to critique its assumptions (e.g., "this is just a profit grab") and instead work *within* its logic to understand its coherence. It required a disciplined act of intellectual empathy to see the world from the perspective of an innovator facing a 15-year, billion-dollar development pathway with no guarantee of recouping costs. Grammar C, while complex, was more aligned with my own critical disposition towards structural inequality, making it easier to simulate.
- **Did you discover blind spots you didn't predict?** Yes. The most significant was the realization of how deeply I had internalized the "excess" framing of consumption (CON1) over the "access" framing. While I was aware of the access problem intellectually, my default processing of rising consumption data was as a stewardship threat. Performing Grammar A forced me to seriously entertain the opposite view: that limiting access to new, effective drugs in an LMIC context could be a form of health injustice. This revealed a normative commitment I hadn't fully articulated.
- **How would the analysis differ from a different home grammar?** If my home grammar were A, I might have struggled more with the seemingly "irrational" behaviour of clinicians who overprescribe, or with the logic of conservation that seems to sabotage the very innovation we need. I might have been blind to the clinical realities that make stewardship difficult. If my home grammar were C, I might have dismissed the concerns of Grammar A as mere corporate lobbying and the concerns of Grammar B as technocratic tinkering, missing the genuine human need for new treatments and the immediate crisis in hospitals.
- **What does this exercise reveal about interpretive architectures?** It reveals that they are not just analytical frameworks, but are deeply intertwined with values, professional identities, and institutional interests. They are self-reinforcing systems. Once you adopt a grammar, it dictates which signals matter, how they are connected, and what counts as a valid solution. The deep-seated conflict in the AMR response is not resolvable by simply presenting "more facts," because the facts themselves are processed into fundamentally different realities. This exercise demonstrates that progress may depend less on finding a single "correct" grammar and more on fostering a meta-dialogue *between* grammars, acknowledging their partiality and seeking policy hybrids that can address the core concerns of each.

SECTION 8: METHODOLOGICAL OBSERVATIONS

The v2.0 of the Prismatic Method represents a significant refinement. The first round (Mark 1) risked producing static, reductive portraits of each grammar. The new features successfully dynamize the analysis.

- **What worked well:**
  - **Counter-signal integration:** Mandating integration *during* processing, rather than deferring it, was crucial. It forced a more honest representation of each grammar, showing not just their internal logic but also the points of tension they must constantly manage. It made the grammars feel more like living, adaptive systems than rigid ideologies.
  - **Temporal orientation:** This was a highly useful parameter. Distinguishing between the forward-looking (A), present-focused (B), and lateral (C) orientations clarified *why* the grammars often talk past each other. They are operating on different time horizons and with different epistemic goals.
  - **Grammar boundaries:** Specifying these was the most insightful addition. It transformed the grammars from static categories into dynamic zones. Seeing how A transforms into B at the clinic door, or B leaks into C at the hospital wall, provides a map for potential communication and coalition-building between interpretive communities.
  - **Stress test:** This was an effective closure mechanism. It moved beyond simply describing the grammars to actively probing their limits and identifying the conditions under which their closures would fail. This adds a crucial layer of analytical humility to the entire exercise.
- **Where did you encounter difficulties?**
  - **Ontological check:** This remains the most challenging but also the most powerful step. Distinguishing between processing the same object differently and processing *different objects* is a subtle but critical analytical move. It requires a high degree of abstraction and was the most difficult section to articulate clearly.
  - **Specifying "Excluded Signals":** It was sometimes difficult to distinguish between a signal a grammar genuinely *excludes* and one it simply *weights very lowly*. For instance, Grammar B doesn't entirely exclude pipeline data, but it makes it nearly invisible. The boundary between low weight and exclusion is fuzzy and needs clearer operationalization in future versions (e.g., "signals that, if presented, would be met with a response of 'that's interesting, but it's not really an AMR issue'").
- **Suggestions for further refinement:**
  - **Power and Asymmetry:** The method currently treats grammars as symmetrical analytical architectures. In reality, they are embedded in fields of power. Grammar A (market) and Grammar B (public health) have vastly more institutional and financial weight than Grammar C (structural). A future iteration (Mark 3) could include a parameter for "institutional weight" or "resource mobilization capacity" to analyze not just how grammars *think*, but how they *act* in the world.
  - **Hybridization and Translation:** Building on "grammar boundaries," the method could be extended to analyze attempts at *translation*—policy documents or interventions that try to speak to multiple grammars. How successful are these hybrids? Do they achieve coherence, or do they paper over fundamental contradictions? This would be a valuable applied extension.